

Supply Chain Complete Analysis

Identifying bottlenecks and profit impact of late deliveries using the DataCo Supply Chain dataset · Python (pandas, NumPy, Matplotlib, Seaborn)

1. Executive Summary

This project analyses 172,765 non-cancelled orders to measure delivery performance, locate the drivers of late delivery, and test whether delays hurt profitability. More than half of all orders (54.7%) ship later than scheduled. **Shipping mode is by far the strongest driver:** First Class orders are almost always late, while region, customer segment, payment type, department, order month, weekday and hour show only marginal differences. Delayed orders earn about the same profit per order as on-time ones, so the cost of delay is more likely to be service-level and customer experience than direct margin.

172,765

Orders analysed

54.7%

Late (94,523 orders)

45.3%

On-time or early

3 days

90th percentile delay (late orders)

2. Business Problem & Desired Outcome

Business problem

A global e-commerce company operating across multiple regions manages end-to-end order fulfilment, including shipping and delivery, for products such as sporting goods. The company faces inconsistent delivery performance: actual shipping times often deviate from scheduled timelines, leading to late deliveries and unpredictable order profitability.

Desired outcome

Analyse delivery operations, identify bottlenecks, and build a predictive system to reduce delays, optimise shipping decisions, and improve overall profitability and efficiency.

This report addresses the analysis and bottleneck-identification goals. The predictive component is outlined under Limitations & Notes.

3. Objectives

- Quantify on-time vs. late delivery performance.
- Detect bottlenecks by region, segment, shipping mode, order status, payment type and department.
- Run a root-cause breakdown for the worst-performing region.
- Test for time-based patterns (month, day of week, hour).
- Assess the relationship between delay and profitability.

4. Data & Methodology

3.1 Dataset

The raw file contains 180,519 rows and 53 columns (orders, customers, products, shipping and profit fields). There were no duplicate rows. Product Description was 100% missing and Order Zipcode about 86% missing (155,679 rows); Customer Lname and Customer Zipcode had 8 and 3 missing values.

3.2 Data cleaning

- Dropped 33 columns: fully missing, redundant (e.g. Benefit per order duplicates Order Profit Per Order), single-valued (Product Status), ID fields, geo-coordinates, and personal data such as customer email, password and name.
- Removed orders with status *Shipping canceled*, leaving **172,765 rows × 26 columns**.
- Converted order and shipping dates to datetime.

3.3 Feature engineering

Feature	Definition
Order Processing Time	Shipping date – order date (days)
Delay	Order Processing Time – scheduled shipment days
Is_Delayed	Delay > 0
order_month / day / hour	Extracted from order date
Profitability Flag	Profit, Break-even or Loss from Order Profit Per Order

5. Key Findings

4.1 Delay distribution

Most late orders are only 1–2 days late. Orders 3–4 days late make up under 8% of the total. The right panel of the figure also shows profit by delay (see 4.5).

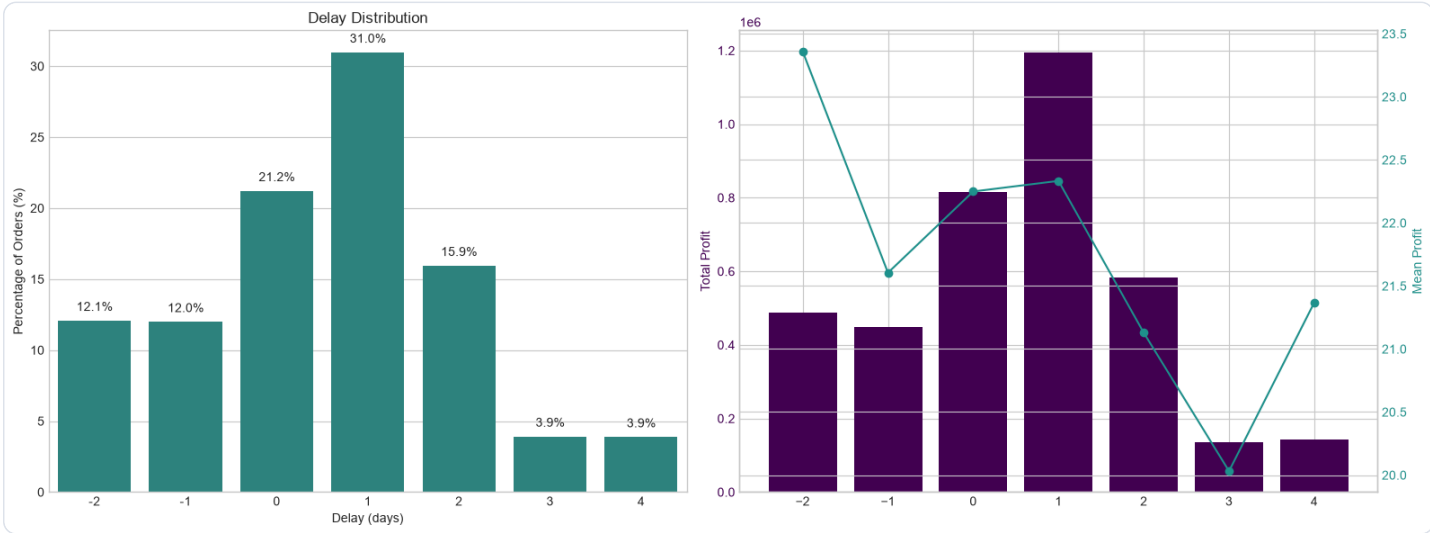


Figure 2: Delay distribution (left) and total / mean profit by delay days (right). Output of the notebook.

4.2 Bottleneck detection

Delay percentage was compared across six dimensions. Only Shipping Mode shows a large spread (shipping-mode values read from the chart, approximate):

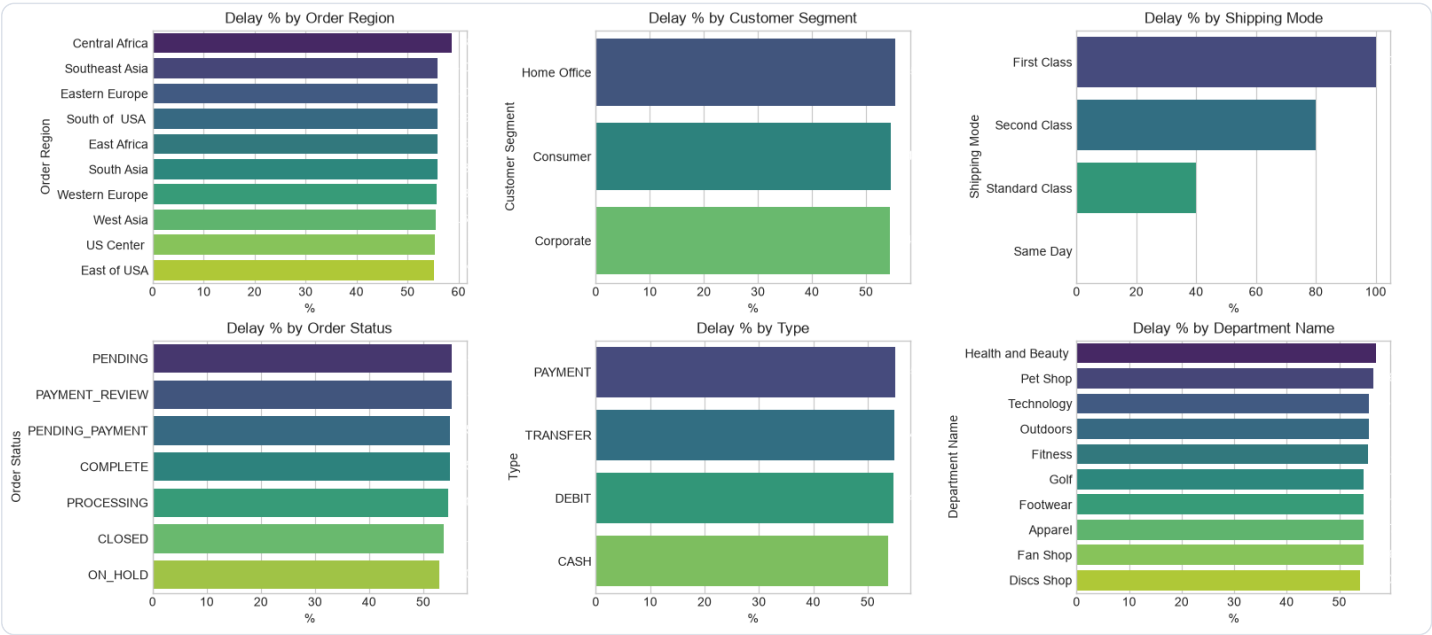


Figure 3: Delay % by region, segment, shipping mode, order status, payment type and department.

Region, customer segment, payment type, order status and department all cluster in a narrow band of roughly 54–58% late. Central Africa is the highest-delay region, but only marginally above the rest.

4.3 Root cause: Central Africa

Within Central Africa, the top drivers were First Class (100%) and Second Class (82.8%) shipping, followed by orders in PAYMENT_REVIEW (80.0%) and PENDING (69.1%) status. Payment type, department and customer segment sit around 60–64%, close to the overall level. This mirrors the global result: shipping mode is the main lever.

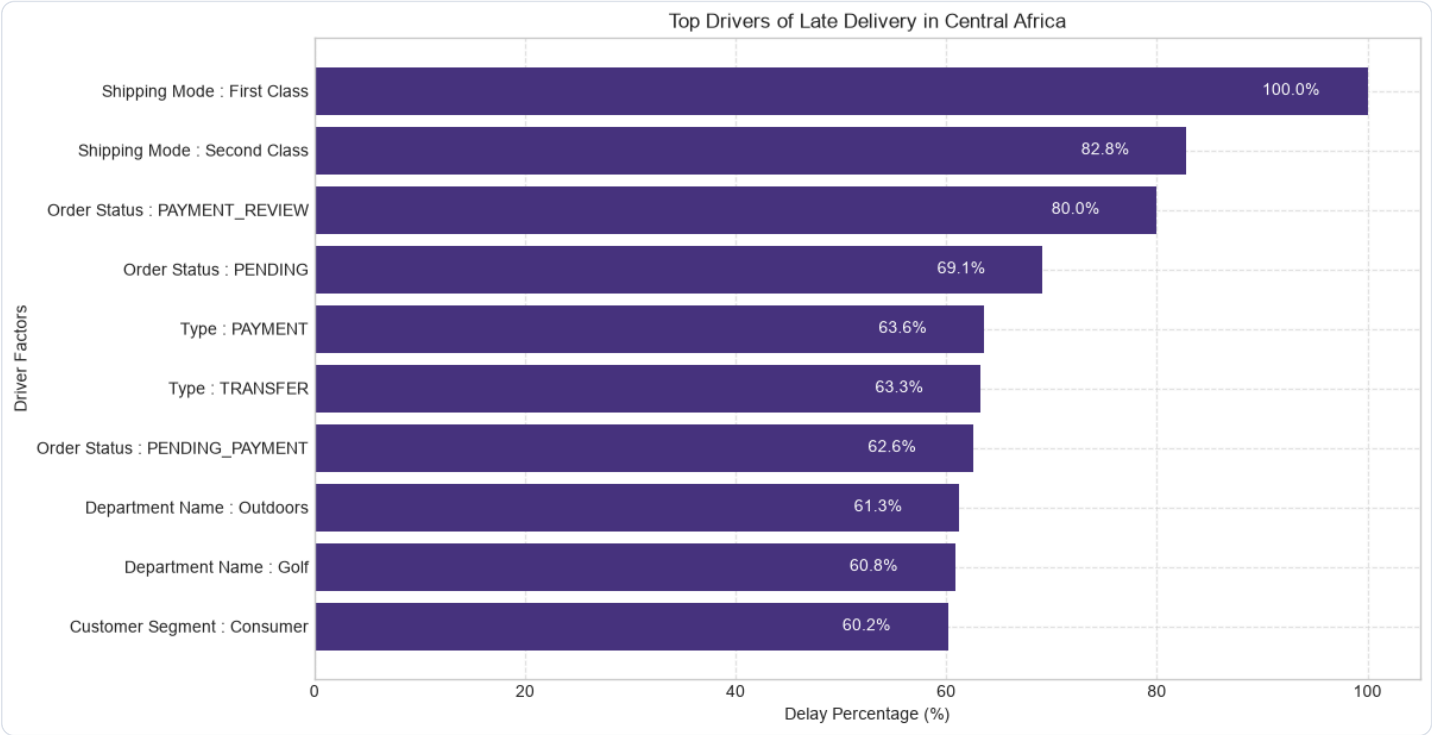


Figure 4: Top drivers of late delivery in Central Africa.

4.4 Time-based patterns

Dimension	Delay % range	Reading
Month	≈53.7% – 55.4%	Peaks Aug/Sep/Dec; lowest in July. Spread under 2 points.
Day of week	≈54.8% – 55.5%	Essentially flat.
Hour	≈51.9% – 57.1%	Widest spread, but noisy with no clear trend.

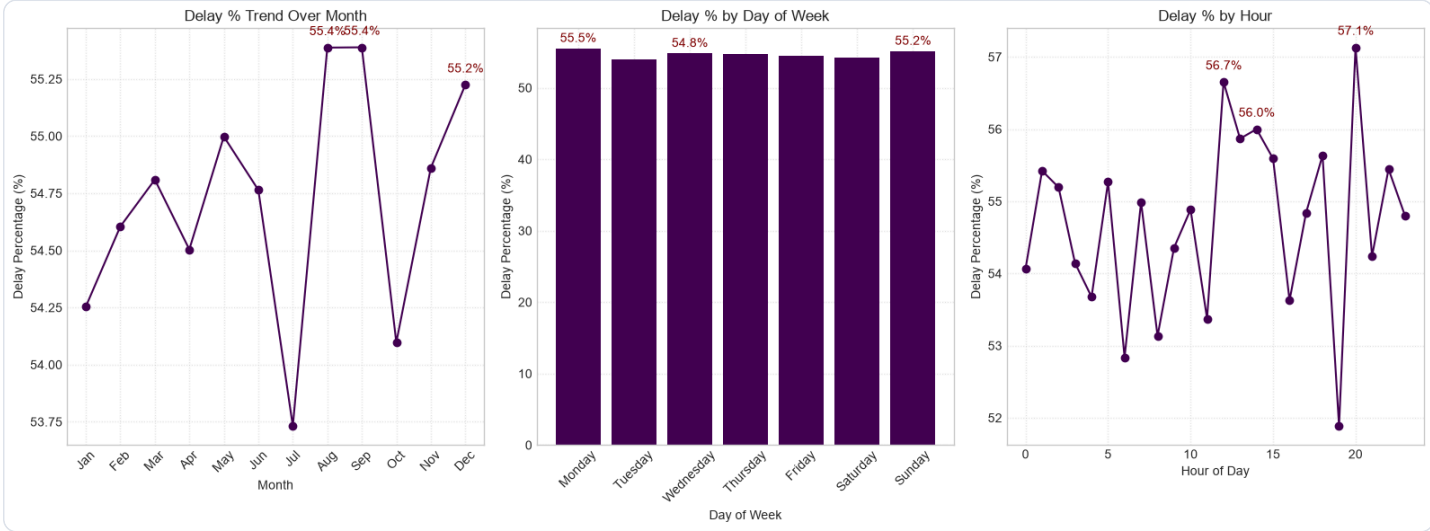
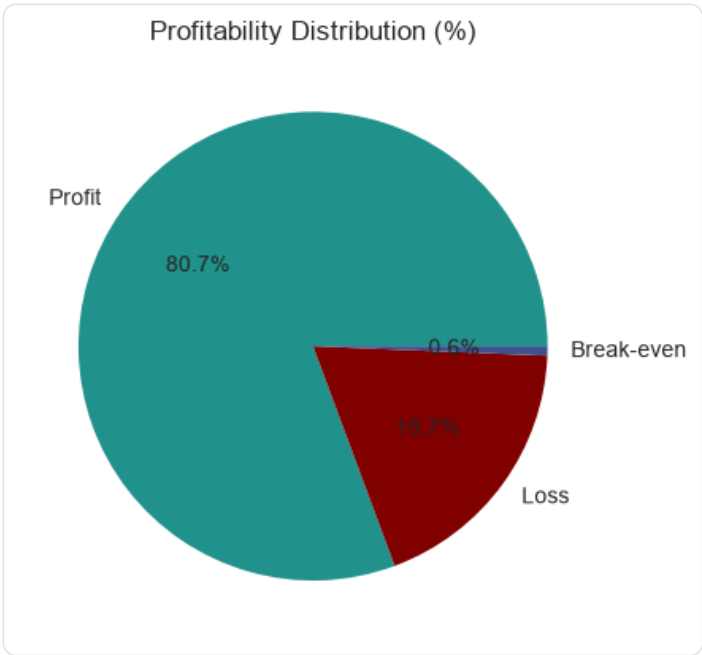


Figure 5: Delay % trend by month, day of week and hour.

Delays are not driven by seasonality or order timing.

4.5 Profitability

Outcome	Orders	Share
Profit	139,354	80.7%
Loss	32,295	18.7%
Break-even	1,116	0.6%



Mean profit per order stays in a narrow band of about 20.0–23.4 across all delay values (–2 to +4 days), with the lowest at +3 days. Total profit peaks at +1 day simply because that group has the most orders (53,503). There is no clear sign that delays reduce per-order profit.

6. Business Impact

First Class orders (26,513) are late almost every time. If that mode had been delivered on schedule, roughly 26.5K of the 94.5K late orders would disappear and the overall late rate would fall from **54.7% to about 39%**, a drop of around 15 points from fixing a single shipping tier. Second Class (33,806 orders, about 80% late) is the next largest opportunity.

7. Recommendations

- **Recalibrate shipping-mode promises.** First and Second Class are late in most cases. Either lengthen the scheduled window to match real fulfilment or fix carrier and warehouse handling for premium tiers.
- **Prioritise premium tiers for monitoring.** Track on-time rate by shipping mode as the primary KPI, since other dimensions add little signal.
- **Speed up orders stuck in review.** PAYMENT_REVIEW and PENDING orders in Central Africa show higher delay rates; shortening review time may help.
- **Investigate loss-making orders separately.** 18.7% of orders lose money regardless of delay, which is a bigger profit issue than lateness.

8. Limitations & Notes

Metric definitions

- **Late delivery** is defined as actual processing days (shipping date – order date) exceeding the scheduled shipment days (Delay > 0), giving 94,523 late orders (54.7%). The dataset's own Delivery Status label marks 98,977 orders (57.3%) as late; the difference of about 2.6 points comes from how each measure is calculated. All results in this report use the Delay > 0 definition.
 - **Profit figures** come from Order Profit Per Order. Profit on delayed orders sums to about 2.1M in the dataset's currency units. The dataset does not state a currency, so no symbol is used here.
 - Total profit summed over profitable orders only is about 7.5M; net profit after losses would be lower.
- The predictive system named in the desired outcome (a model to forecast late deliveries) is not part of this version of the analysis and is the natural next step.
 - The analysis is descriptive. Differences between groups are shown as rates and were not tested for statistical significance.
 - Shipping mode may be confounded with scheduled days (First Class is scheduled for 1 day), so high late rates may reflect tight promises rather than poor operations.
 - Cancelled orders were excluded, which may understate operational problems.

9. Tools

Python, pandas, NumPy, Matplotlib and Seaborn in Jupyter Notebook.

Source: DataCo Supply Chain Dataset · Report generated from the accompanying Jupyter notebook analysis.